

Department of Electronics and Electrical Engineering
Indian Institute of Technology Guwahati

EE101 - Basic Electronics

Pre-Tutorial Assignment 5: Solutions

1. A linear network **N** (with unique solution) delivers $3.5A$ when the load connected is 5.25Ω and $2.1A$ when the load is 19.5Ω . What is the maximum power $P_{max}(\mathbf{N})$ (in W) that **N** can deliver?

Solution

The linear network **N** has a unique solution and thus, it can be represented by its Thevenin's equivalent network (voltage source V_{OC} in series with resistance R_{TH}) with the load connected across the terminals $X - Y$ (Figure 1).

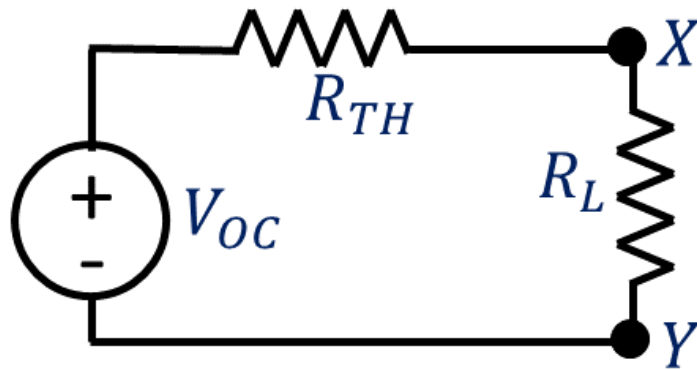


Figure 1: The Thevenin's equivalent circuit for the network **N** with the load R_L connected across terminals $X - Y$.

The network delivers $3.5A$ when $R_L = 5.25\Omega$. Thus, we can write

$$V_{OC} = 3.5 \times (R_{TH} + 5.25) \quad (1)$$

Again, the network delivers $2.1A$ when $R_L = 19.5\Omega$. Thus, we can write

$$V_{OC} = 2.1 \times (R_{TH} + 19.5) \quad (2)$$

Solving Equations 1 and 2, we get $R_{TH} = 16.125\Omega$ and $V_{OC} = 74.8125V$. Thus, the maximum power delivered by **N** is

$$P_{max}(\mathbf{N}) = \frac{V_{OC}^2}{R_{TH}} = 86.7738W \quad (3)$$

2. Consider an Excess-3 to BCD code converter. Let the Excess 3 input be represented by four bits $PQRS$ and the BCD output be represented by four bits $ABCD$.

- Draw the K-maps for A and B
- Find a Boolean expression for A in the minimal SOP form.
- Find a Boolean expression for B in the minimal POS form.

Solution

The truth table relating the inputs to the outputs can be seen as below:

Decimal	P	Q	R	S	A	B	C	D
0	0	0	1	1	0	0	0	0
1	0	1	0	0	0	0	0	1
2	0	1	0	1	0	0	1	0
3	0	1	1	0	0	0	1	1
4	0	1	1	1	0	1	0	0
5	1	0	0	0	0	1	0	1
6	1	0	0	1	0	1	1	0
7	1	0	1	0	0	1	1	1
8	1	0	1	1	1	0	0	0
9	1	1	0	0	1	0	0	1

		RS			
		00	01	11	10
PQ	00	ϕ	ϕ	0	ϕ
	01	0	0	0	0
	11	1	ϕ	ϕ	ϕ
	10	0	0	1	0

The K-map for A is:

The minimal SOP for A is: $A = P.Q + P.R.S$

		RS			
		00	01	11	10
PQ	00	ϕ	ϕ	0	ϕ
	01	0	0	1	0
	11	0	ϕ	ϕ	ϕ
	10	1	1	0	1

The K-map for B is:

The minimal POS for B is: $B = (P + S).(\overline{Q} + R).(Q + \overline{R} + \overline{S})$ and $B = (P + R).(\overline{Q} + S).(Q + \overline{R} + \overline{S})$